

Optimal Control of a Variable-Mass Thrust-Vectoring Lander*

Nathan Ge^{1‡}

Abstract

This project studies the optimal guidance and control of a planar variable-mass rocket/lander with translational and rotational dynamics, bounded controls, nonlinear coupling, free-final-time optimal control, endpoint manifold constraints, and trajectory tracking. The project combines Pontryagin's Maximum Principle (PMP), controllability analysis, Linear Quadratic Regulation (LQR), minimum-time optimal control, nonlinear indirect optimal control, and numerical two-point boundary value problem (TPBVP) solvers. The project is inspired by powered-descent guidance and reusable rocket landing systems studied in modern aerospace optimal control literature. All theoretical derivations must be based on PMP. Numerical implementations may use indirect shooting, multiple shooting, continuation or homotopy methods, and indirect/direct hybrid approaches, provided that the optimality conditions are derived from PMP.

github repository:
link<https://github.com/nzge/optimal-control-lander>

1. Problem Formulation

The state space of the lander can be represented as:

$$x = [p_x \ p_z \ v_x \ v_z \ \theta \ \omega \ m]^\top, \quad u = [\delta T \ \tau]^\top$$

Symbol	Description
<i>State Variables (x)</i>	
p_x, p_z	Horizontal and vertical position
v_x, v_z	Horizontal and vertical velocity
θ	Vehicle pitch angle
ω	Angular velocity (pitch rate)
m	Instantaneous vehicle mass
<i>Control Variables (u)</i>	
T	Commanded thrust magnitude
τ	Commanded control torque

The nonlinear equations of motion governing the lander, $\dot{x} = f(x, u)$, are defined by its kinematics, gravity, thrust resolution, and mass depletion rate.

$$\dot{x} = \begin{bmatrix} v_x \\ v_z \\ \frac{T}{m} \sin(\theta) \\ \frac{T}{m} \cos(\theta) - g \\ \omega \\ \frac{\tau}{I_{sp} g_0} \\ -\frac{\tau}{I_{sp} g_0} \end{bmatrix}$$

Table 1: System Constraints

Mathematical Constraint	Description
<i>Control Constraints</i>	
$T \in [T_{\min}, T_{\max}]$	Actuator bounds on thrust magnitude
$\tau \in [\tau_{\min}, \tau_{\max}]$	Actuator bounds on control torque
<i>Numerically Enforced State Constraints</i>	
$p_z(t) \geq 0$	Hard deck limit preventing subsurface flight
$m(t) \geq m_{\text{dry}}$	Minimum mass limit (dry mass)

*This work was not supported by any organization

[†]Nathan Ge is a student in the Department of Mechanical and Aerospace Engineering, University of California, Los Angeles, 405 Hilgard Avenue, Los Angeles, CA 90095 nzge@g.ucla.edu

Table 2: System Parameters

Symbol	Value	Description
g	9.81 m/s^2	Gravitational acceleration
I	$10 \text{ kg} \cdot \text{m}^2$	Vehicle moment of inertia
m_0	20 kg	Initial vehicle mass
α	$0.005 \text{ kg}/(\text{N} \cdot \text{s})$	Propellant consumption rate
$T_{\min}$	0 N	Minimum thrust limit
$T_{\max}$	300 N	Maximum thrust limit
$\tau_{\min}$	$-30 \text{ N} \cdot \text{m}$	Minimum torque limit
$\tau_{\max}$	$30 \text{ N} \cdot \text{m}$	Maximum torque limit

2. Numerical Methodology

2.1. Linear Systems Solver: The Sweep Method (Parts I & II)

`scipy.integrate.solve_ivp` integrates the Riccati equation backward in time to generate the feedback gain matrix $K(t)$. An interpolator was then used on this gain during the forward-pass simulation of the state trajectory.

2.2. Nonlinear TPBVP Solvers (Parts III & IV)

The Riccati substitution no longer applies once the optimal control problem is nonlinear, so the coupled state/costate system arising from PMP is solved directly. Part III uses *indirect single shooting* on the free-final-time mission: the $2n$ canonical equations are integrated forward and a Newton iteration adjusts the unknown initial costate, the phase-switch time, and the final time until the manifold, transversality, and $H(t_f) = 0$ residuals vanish. Part IV uses *indirect collocation* (`scipy.integrate.solve_bvp`) on the fixed-time problem: the 14 state/costate equations are discretized on a mesh and the optimal control is substituted in closed form from the stationarity conditions. Collocation is preferred for Part IV because it accepts a full state/costate *trajectory* as its initial guess, which is taken from the corresponding linearized LQR solution; this warm start lies close to the nonlinear extremal and lets the solver converge in a few mesh-refinement iterations. In both parts the actuator bounds are imposed by saturating the PMP-optimal control, consistent with Pontryagin's pointwise minimization of H over the admissible control set.

2.3. Constraint Enforcement Strategies

Control Constraints (T, τ): For the linear LQR phases I applied post-calculation saturation, clipping the commanded δT and τ to $[T_{\min}, T_{\max}]$ and $[\tau_{\min}, \tau_{\max}]$. For the nonlinear PMP phases the control follows in closed form from the stationarity condition $\partial H / \partial u = 0$ and is then clipped to the same actuator bounds inside the solver right-hand side; this is exactly Pontryagin's pointwise minimization of H , since for the control-affine Hamiltonian the constrained minimizer is the unconstrained stationary control projected onto the admissible box.

State Constraints ($p_z \geq 0, m \geq m_{dry}$): Adjoining state constraints directly to the PMP Hamiltonian creates severe analytical difficulties (jump discontinuities in the costates). The numerical workaround is to either:

"Soft Constraints" (adding massive penalties to your Q matrix if the rocket goes underground)

"Event Tracking" (terminating the ODE solver upon impact)

Direct constraint arrays in an NLP solver.

3. Part I — Fixed-Time LQR Around Different Nominal Trajectories

3.1. Phase Formulation

Linearization: The linearized model plant is formulated as $\dot{x} = Ax + Bu$ where $A = \frac{\partial f}{\partial x}$, $B = \frac{\partial f}{\partial u}$, and $f(x, u) = \dot{x}$.

Hover

$$A_{\text{hover}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{g}{m_0} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B_{\text{hover}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_0} & 0 \\ 0 & 0 \\ 0 & I^{-1} \\ -\alpha & 0 \end{bmatrix}$$

Descent

$$A(t) = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{g}{m^*(t)} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B(t) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m^*(t)} & 0 \\ 0 & 0 \\ 0 & 1/I \\ -\alpha & 0 \end{bmatrix}$$

$$m^*(t) = m_0 e^{-\alpha g t}$$

3.2. Analytical Derivations (PMP)

Hamiltonian construction

$$H(x, u, p) = L(x, u) + p^\top f(x, u)$$

$$\text{where } L(x, u) = x^\top Qx + u^\top Ru$$

$$H(x, u, p) = x^\top Qx + u^\top Ru + p^\top (Ax + Bu)$$

Transversality

$$p(t_f) = \left. \frac{\partial \phi}{\partial x} \right|_{t=t_f}$$

$$p(t_f) = 2Q_f x(t_f)$$

Adjoint Equations (costate dynamics)

$$\dot{p} = -\frac{\partial H}{\partial x}$$

Using matrix calculus (and the symmetry of Q):

$$\dot{p} = -(2Qx + A^\top p) = -2Qx - A^\top p$$

Stationarity Conditions

$$\frac{\partial H}{\partial u} = 0 \implies 2Ru + B^\top p = 0 \implies u^* = -\frac{1}{2}R^{-1}B^\top p$$

Riccati Differential equation Assumption (Riccati Substitution):

$$p(t) = P(t)x(t)$$

Where $P(t)$ is an unknown, time-varying symmetric matrix. Differentiate the assumption using the product rule:

$$\dot{p} = \dot{P}x + P\dot{x}$$

Substitute $\dot{p}$ and $\dot{x}$ (the dynamics).

$$-2Qx - A^\top Px = \dot{P}x + P(Ax + Bu^*)$$

Substitute the optimal control u^* . Remembering that $p = Px$, it becomes $u^* = -\frac{1}{2}R^{-1}B^\top Px$.

$$-2Qx - A^\top Px = \dot{P}x + P\left(Ax - \frac{1}{2}BR^{-1}B^\top Px\right)$$

$$-2Qx - A^\top Px = \dot{P}x + PAx - \frac{1}{2}PBR^{-1}B^\top Px$$

Group all terms multiplying $x(t)$. Move everything to one side of the equation:

$$\left[\dot{P} + PA + A^\top P - \frac{1}{2}PBR^{-1}B^\top P + 2Q \right] x = 0$$

Because this equation must hold true for any arbitrary state trajectory $x(t)$, the bracketed matrix expression must identically equal zero. Setting the bracket to zero yields the Riccati Differential Equation (RDE):

$$\dot{P} = -PA - A^\top P + \frac{1}{2}PBR^{-1}B^\top P - 2Q$$

Optimal State-feedback law

$$P(t_f) = 2Q_f, \quad u^*(t) = -K(t)x(t), \quad K(t) = \frac{1}{2}R^{-1}B^\top P(t)$$

3.3. Numerical Implementation and LQR Tuning

Q , R and Q_f matrix comparisons. We used heuristic methods to create a selection of Q , R , and Q_f that accurately reflect underlying control intuition.

Table 3: Part I.5.5 cost presets (diagonal Q , R , Q_f).

Preset	$Q = \text{diag}(\cdot)$	R	$Q_f = \text{diag}(\cdot)$
balanced	(1, 1, 0.5, 0.5, 10, 5, 0.01)	(0.1, 1)	(10, 10, 1, 1, 50, 20, 0.1)
pos_heavy	(20, 20, 2, 2, 5, 2, 0.01)	(0.1, 1)	(50, 50, 5, 5, 20, 10, 0.1)
att_heavy	(1, 1, 0.5, 0.5, 50, 25, 0.01)	(0.1, 2)	(10, 10, 1, 1, 100, 50, 0.1)
cheap_th	(1, 1, 0.5, 0.5, 10, 5, 0.01)	(0.01, 1)	(10, 10, 1, 1, 50, 20, 0.1)
exp_trq	(1, 1, 0.5, 0.5, 10, 5, 0.01)	(0.1, 10)	(10, 10, 1, 1, 50, 20, 0.1)
scaled_id	(10^{-2} , 10^{-2} , 0.04, 0.04, 25, 4, 1)	(1, 1)	same as Q

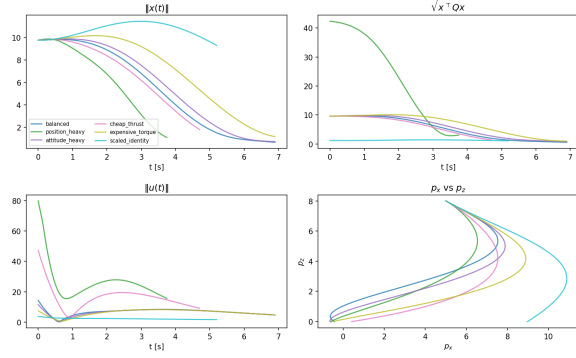


Figure 1: Cost sweep

3.4. Analysis

Controllability Analysis

Algebraic Controllability Matrix (Frozen-Time):
Used to check the structural control authority of the LTI system (or a "frozen" snapshot of an LTV system). For a system with $n = 7$ states:

$$C = [B \ AB \ A^2B \ A^3B \ A^4B \ A^5B \ A^6B]$$

Finite-Horizon Controllability Gramian:

$$W_c(0, t_f) = \int_0^{t_f} \Phi(0, \tau) B(\tau) B(\tau)^\top \Phi(0, \tau)^\top d\tau$$

Where $\Phi(t, \tau)$ is the state transition matrix of $A(t)$.

Numerical Computation (Differential Equation):
To practically compute W_c numerically and avoid explicitly calculating the integral and state transition matrix, we can solve its differential equation form:

$$\dot{W}_c(t) = A(t)W_c(t) + W_c(t)A(t)^\top + B(t)B(t)^\top$$

Minimum Control Energy: The minimum control energy E required to reach a specific state deviation x_f is given by:

$$E = x_f^\top W_c^{-1} x_f$$

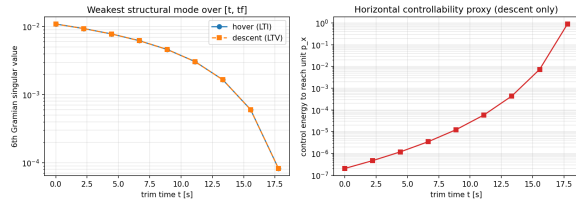


Figure 2: Controllability Gramian and horizontal controllability proxy

Controllable energy in the weakest meaningful mode shrinks as $t_f - t$ shrinks, especially for horizontal motion (as shown by the proxy).

Controllability matrices

State order: $x = [p_x, p_z, v_x, v_z, \theta, \omega, m]^\top$

controls $u = [\delta T, \tau]^\top$

$\mathcal{C} = [B, AB, \dots, A^6B] \in \mathbb{R}^{7 \times 14}$.

Trim Snapshot $t = 0.0s$

Descent trim: $m = 20.00kg$, $v_x^* = 40.0m/s$,
 $v_z^* = -8.0m/s$.

Table 4: Controllability rank and singular values

Model	rank($\mathcal{C}$)	σ_1	σ_6	σ_7
Hover LTI	7/7	0.98	0.0489	0
Descent LTV	7/7	0.98	0.0489	0

Hover LTI at $t = 0.0s$.

	B	AB	A^2B	A^3B	A^4B	A^5B	A^6B
p_x	0	0	0	0	0	0.98	0
p_z	0	0	0.05	0	0.00245	0	0
v_x	0	0	0	0	0.98	0	0
v_z	0.05	0	0.00245	0	0	0	0
θ	0	0	0	0.1	0	0	0
ω	0	0.1	0	0	0	0	0
m	-0.005	0	0	0	0	0	0

Descent LTV at $t = 0.0s$.

	B	AB	A^2B	A^3B	A^4B	A^5B	A^6B
p_x	0	0	0	0	0	0.98	0
p_z	0	0	0.05	0	0.00245	0	0
v_x	0	0	0	0	0.98	0	0
v_z	0.05	0	0.00245	0	0	0	0
θ	0	0	0	0.1	0	0	0
ω	0	0.1	0	0	0	0	0
m	-0.005	0	0	0	0	0	0

Trim Snapshot $t = 15.0s$

Descent trim: $m = 9.59kg$, $v_x^* = 25.0m/s$,
 $v_z^* = -8.0m/s$.

Table 5: Controllability rank and singular values

Model	rank($\mathcal{C}$)	σ_1	σ_6	σ_7
Hover LTI	7/7	0.98	0.0489	0
Descent LTV	7/7	0.98	0.1	0

Hover LTI at $t = 15.0s$.

	B	AB	A^2B	A^3B	A^4B	A^5B	A^6B
p_x	0	0	0	0	0	0.98	0
p_z	0	0	0.05	0	0.00245	0	0
v_x	0	0	0	0	0.98	0	0
v_z	0.05	0	0.00245	0	0	0	0
θ	0	0	0	0.1	0	0	0
ω	0	0.1	0	0	0	0	0
m	-0.005	0	0	0	0	0	0

Descent LTV at $t = 15.0s$.

	B	AB	A^2B	A^3B	A^4B	A^5B	A^6B
p_x	0	0	0	0	0	0.98	0
p_z	0	0	0.104	0	0.00511	0	0
v_x	0	0	0	0	0.98	0	0
v_z	0.104	0	0.00511	0	0	0	0
θ	0	0	0	0.1	0	0	0
ω	0	0.1	0	0	0	0	0
m	-0.005	0	0	0	0	0	0

Controllable and weakly controllable linearization comparisons.

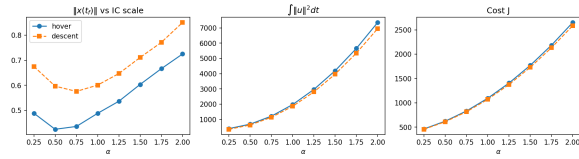


Figure 3: Sensitivity to initial condition error

At small perturbations: descent can be worse (weak plant + wrong linearization point). At large perturbations: hover overtakes — hover LTI may be overly optimistic at large α (linear model stays fixed while state explores larger deviations).

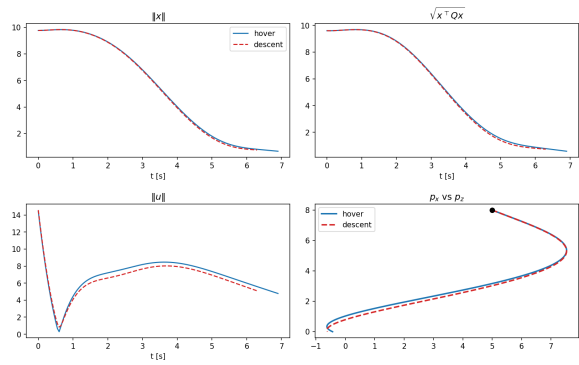


Figure 4: State and control effort across trajectory

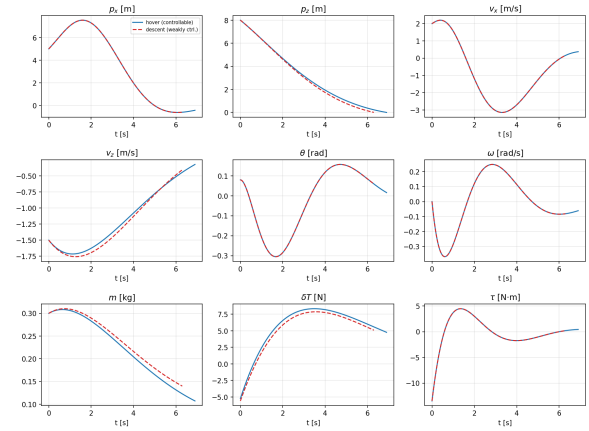


Figure 6: Optimal solution trajectories versus time: states and control histories.

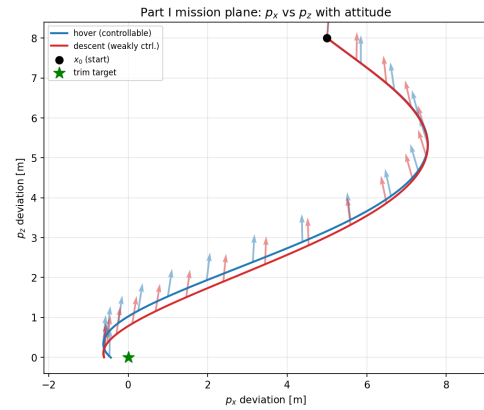


Figure 7: Mission-plane trajectory (p_x vs p_z) with attitude vectors at sampled positions.

3.5. Validation

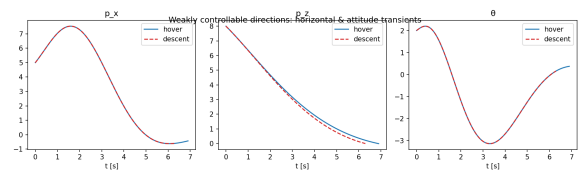


Figure 5: Transient behavior

p_x and θ swings indicate structural difficulty. Even “controllable” hover struggles. Gramian analysis flagged as expensive to correct. The gap is clearest in p_z (mass/thrust schedule, changing trim). Descent worse

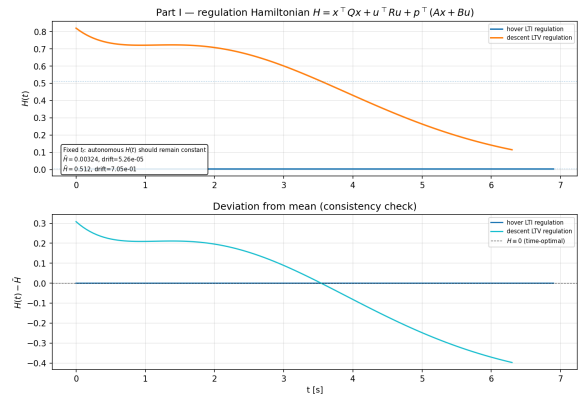


Figure 8: Hamiltonian evolution

Because this phase utilizes the Riccati sweep method rather than a shooting solver, optimality is validated by verifying the backward-pass terminal conditions and forward-pass tracking convergence. Table 6 confirms the Riccati matrix $P(t)$ strictly satisfies the terminal boundary condition $P(t_f) = Q_f$. Additionally, the closed-loop tracking error $e(t)$ is bounded within acceptable numerical tolerances at the final time.

Table 6: Riccati and Tracking Validation (Phase I/II)

Metric	Target	Computed	Error (Residual)
$\ P(t_f) - Q_f\ _\infty$	0	1.22×10^{-14}	1.22×10^{-14}
$\ e(t_f)\ _\infty$	0	4.51×10^{-6}	4.51×10^{-6}

4. PART II - Optimal Trajectory Tracking

4.1. Phase formulation

Objective (free final state, no terminal cost, as specified for Part II):

$$J = \int_0^{t_f} \left[(x - x_{\text{ref}})^\top Q (x - x_{\text{ref}}) + u^\top R u \right] dt$$

4.2. Scenario and Operating Point

The reference $x_{\text{ref}}(t)$ is one of the optimal regulation trajectories recorded in Part I (Fig. 10): a damped maneuver that settles to the hover trim. To produce a nontrivial transient, the controller is initialized at a *substantially different* state, $x_0^{\text{trk}} = [-12, 15, -4, 3, -0.25, 0.4, -0.8]^\top$, deviating from the reference start by $\Delta p_x \approx 17$ m and $\Delta \theta \approx 0.33$ rad. Because the hover linearization is altitude-invariant (p_z is decoupled and absent from every right-hand side), the maneuver is executed about a hover point $h = 12$ m above the pad; this leaves A , B , x_{ref} , and the feedback law unchanged while the absolute altitude $p_z = h + p_z^{\text{dev}}$ is used to enforce the ground constraint $p_z \geq 0$. The numerically enforced state constraints $p_z \geq 0$ and $m \geq m_{\text{dry}}$ are satisfied throughout: the minimum absolute altitude is 11.6 m (reference) and 10.0 m (tracker), and the mass margin remains above 12 kg.

4.3. Analytical Derivations (PMP)

The Hamiltonian:

$$H = (x - x_{\text{ref}})^\top Q (x - x_{\text{ref}}) + u^\top R u + p^\top (Ax + Bu)$$

Adjoint Equation (Costate Dynamics):

$$\begin{aligned} \dot{p} &= -\frac{\partial H}{\partial x} \\ \dot{p} &= -2Q(x - x_{\text{ref}}) - A^\top p \end{aligned}$$

Stationarity (Optimal Control Law):

$$\frac{\partial H}{\partial u} = 0 \implies 2Ru + B^\top p = 0 \implies u^* = -\frac{1}{2}R^{-1}B^\top p$$

Transversality (Terminal Condition):

With a free final state and no terminal cost ($\phi \equiv 0$), the transversality condition gives a homogeneous terminal costate:

$$p(t_f) = \frac{\partial \phi}{\partial x} \Big|_{t=t_f} = 0$$

Tracking Riccati Equations (Riccati substitution)

The added auxiliary vector $s(t)$ accounts for the tracking offset:

$$p(t) = P(t)x(t) + s(t)$$

$$\dot{p} = \dot{P}x + P\dot{x} + \dot{s}$$

Substitute $\dot{p}$ and $\dot{x}$:

$$-2Q(x - x_{\text{ref}}) - A^\top(Px + s) = \dot{P}x + P(Ax + Bu^*) + \dot{s}$$

Substitute $u(u^* = -\frac{1}{2}R^{-1}B^\top(Px + s))$:

$$\begin{aligned} -2Qx + 2Qx_{\text{ref}} - A^\top Px - A^\top s = \\ \dot{P}x + PAx - \frac{1}{2}PBR^{-1}B^\top(Px + s) + \dot{s} \end{aligned}$$

Group the terms by $x(t)$:

$$\begin{aligned} \left[\dot{P} + PA + A^\top P - \frac{1}{2}PBR^{-1}B^\top P + 2Q \right] x(t) \\ + \left[\dot{s} + A^\top s - \frac{1}{2}PBR^{-1}B^\top s - 2Qx_{\text{ref}} \right] = 0 \end{aligned}$$

Since the expression must hold true for any arbitrary trajectory $x(t)$, we get the following tracking Riccati pair:

Matrix Riccati Differential Equation (The Feedback Rulebook):

$$\dot{P} = -PA - A^\top P + \frac{1}{2}PBR^{-1}B^\top P - 2Q$$

Auxiliary Vector Differential Equation (The Feedforward Tracker):

$$\dot{s} = -\left(A^\top - \frac{1}{2}PBR^{-1}B^\top\right)s + 2Qx_{\text{ref}}$$

Since $p(t_f) = P(t_f)x(t_f) + s(t_f) = 0$ must hold for an arbitrary terminal state, both pieces are integrated backward from the homogeneous terminal conditions

$$P(t_f) = 0, \quad s(t_f) = 0,$$

reflecting the absence of any terminal penalty in the Part II objective.

Closed-Loop Tracking Control Law

Once you integrate both $\dot{P}$ and $\dot{s}$ backward from t_f to 0, you can construct your final control law. Substitute $p(t) = P(t)x(t) + s(t)$ back into u :

$$u^*(t) = -\frac{1}{2}R^{-1}B^\top P(t)x(t) - \frac{1}{2}R^{-1}B^\top s(t)$$

This is elegantly split into two components:

$$u^*(t) = -K(t)x(t) + u_{ff}(t)$$

$K(t)x(t)$ is the feedback component that stabilizes the rocket and kills deviations. $u_{ff}(t)$ is the feedforward component that anticipates the control effort.

4.4. Analysis

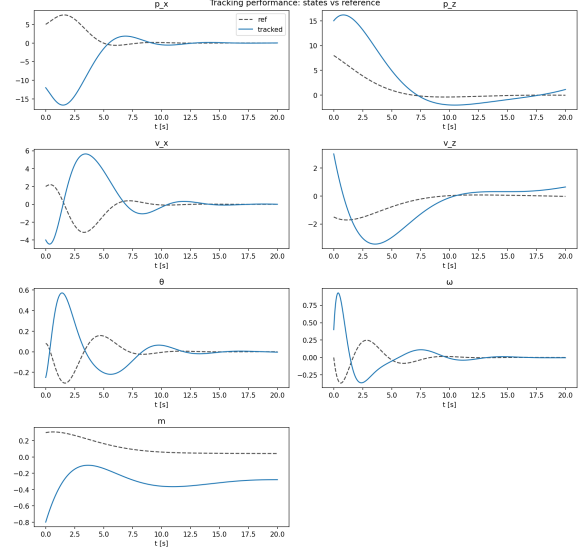


Figure 9: Tracking Performance

Tracker rejoins the reference qualitatively. Errors shrink and shapes align by mid-horizon.

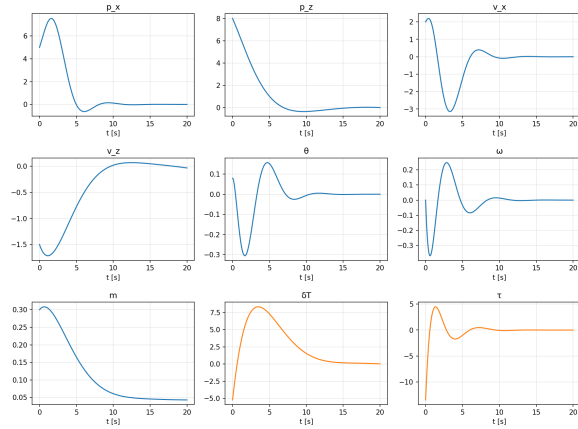


Figure 10: Recorded Part I optimal trajectory used as the reference $x_{\text{ref}}(t)$, $u_{\text{ref}}(t)$.

Quantitatively, the tracker starts with state error $\|e(0)\| \approx 19.9$, peaks at $\|e\| \approx 26.2$ near $t = 1.6$ s as it first accelerates toward the reference, and the cost-weighted error decays to within 10% of its peak by $t \approx 6.9$ s. The residual terminal error $\|e(t_f)\| \approx 1.38$ is concentrated in the vertical and mass channels ($|e_{p_z}| \approx 1.16$, $|e_{v_z}| \approx 0.67$, $|e_m| \approx 0.32$); the horizontal and attitude errors are driven to $\mathcal{O}(10^{-3})$. This residual is an expected feature of the fixed-horizon, free final-state problem with no terminal cost ($P(t_f) = 0$): as $t \rightarrow t_f$ the

Riccati gain $K(t)$ and feedforward $s(t)$ both decay, so the controller “relaxes” near t_f and the weakly penalized vertical/mass states drift slightly.

The required mission-plane views (Figs. 11–12) plot p_x versus p_z with body-axis attitude arrows. The tracker spirals down from its displaced initial state and converges onto the reference at the hover target, remaining well above the ground line throughout.

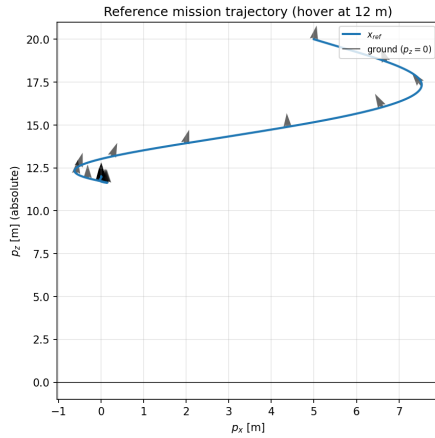


Figure 11: Reference mission trajectory in the p_x - p_z plane with attitude.

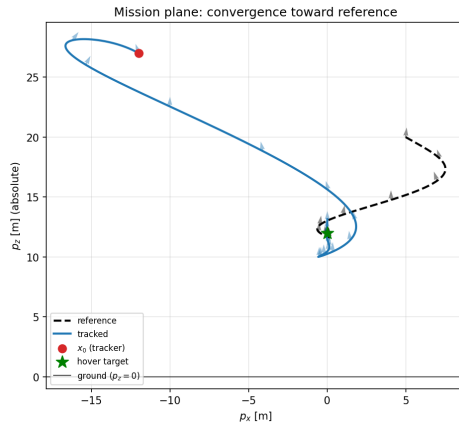


Figure 12: Closed-loop convergence of the tracker onto the reference, with attitude arrows and the ground constraint $p_z = 0$.

Transient Response

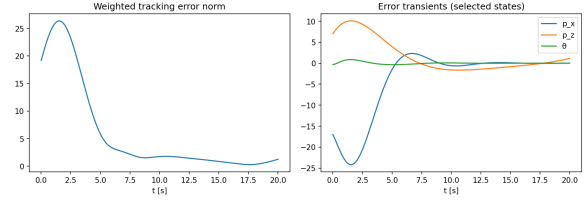


Figure 13: Transient Response

Scalar tracking error (cost-balanced error) decays monotonically in the weighted norm. Moving target, not zero, so no rebound mid-horizon. Position errors decay slowly and dominate; attitude error stays small throughout. Playing ‘catch-up’.

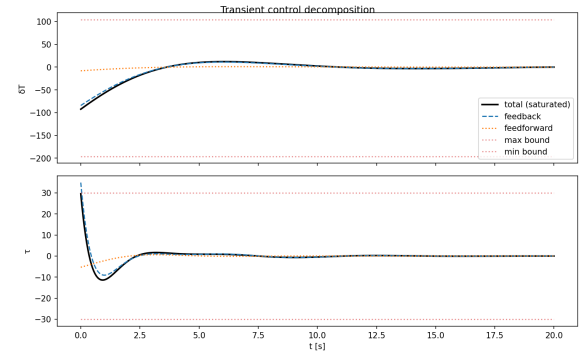


Figure 14: Transient Control Decomposition

The control splits as $u^*(t) = -K(t)e(t) + u_{ff}(t)$. The feedback term dominates the transient and drives the large initial error to zero, while the feedforward term $u_{ff} = -R^{-1}B^T s$ is small and slowly varying because the reference is itself a decaying signal. The torque channel uses nearly the full authority at $t = 0$ ($\tau \approx 29.6$ N·m against the ± 30 N·m bound) to arrest the attitude error, but no channel saturates over the horizon.

Control Effort

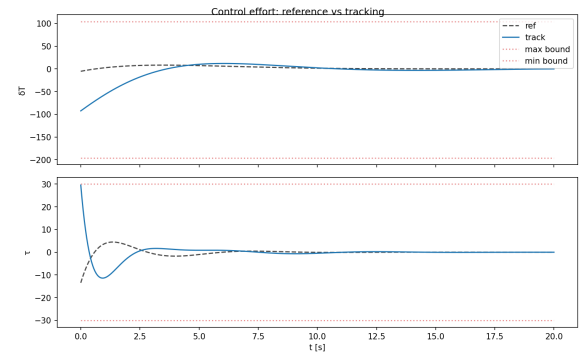


Figure 15: Control Effort

Relative to the smooth reference plan, rejoining from the displaced initial state is expensive: the tracker expends $\int u^\top R u dt \approx 1041$ versus ≈ 86 for the reference, a $12\times$ increase concentrated in the first few seconds. Peak thrust deviation reaches $\delta T \approx 92$ N (within the $+104$ N headroom to $T_{\max}$), confirming that the chosen weights keep the catch-up maneuver feasible without thrust saturation. Once the error is nulled, the tracking control collapses onto the reference control.

Robustness to Deviations from the Nominal Trajectory

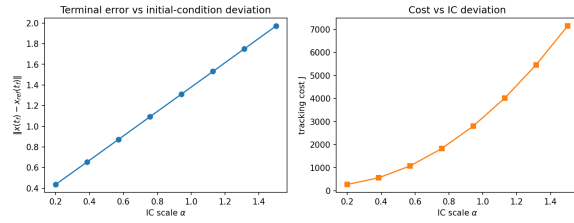


Figure 16: Robustness to initial-condition deviations: terminal error and tracking cost as the initial offset is scaled by α .

Sweeping the initial offset by a factor α confirms the structure of the linear tracking law: the terminal error grows essentially *linearly* in α (slope ≈ 1.18), consistent with superposition of the closed-loop transition map, while the tracking cost grows *quadratically* (the cost is a quadratic form in the deviation). There is no threshold or instability over the tested range, demonstrating graceful degradation with respect to initial-condition error.

Robustness to errors in the assumed reference is examined in Fig. 17, where the reference used to build the feedforward is corrupted by zero-mean noise of level σ and the resulting controller is scored against the *true* reference. The terminal error is only mildly affected (from 1.38 at $\sigma = 0$ to 1.50 at $\sigma = 0.30$) because the feedback term rejects high-frequency reference error, but the tracking cost degrades markedly (from ≈ 3150 to ≈ 6810 , a $2.2\times$ increase) as the controller wastes effort chasing the corrupted plan.

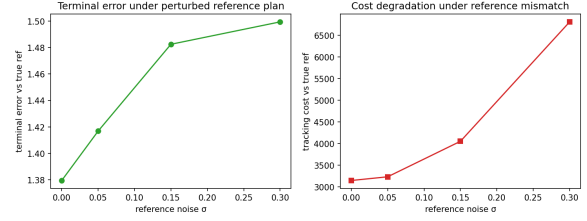


Figure 17: Sensitivity to a perturbed reference plan: terminal error and cost (scored against the true reference) versus reference noise level σ .

4.5. Discussion: Unknown Reference Trajectory

What if the reference trajectory is not known in advance? The present design is fundamentally *non-causal*: the feedforward $s(t)$ is obtained by integrating $\dot{s} = -(A^\top - PBR^{-1}B^\top)s + Qx_{\text{ref}}$ backward from $s(t_f) = 0$, which requires the entire future reference $\{x_{\text{ref}}(\tau) : \tau \in [t, t_f]\}$ to be available at the current time t . Three challenges follow directly:

Loss of anticipation. If only the present (or causally measured) reference is known, $s(t)$ cannot be computed and the controller collapses to pure feedback $u = -K(t)e(t)$. This still stabilizes the error dynamics, but it lags a time-varying reference and incurs larger transient and steady tracking error, since the anticipatory term that pre-positions the control is gone.

Sensitivity to reference model error. As Fig. 17 shows, the feedforward is only as good as the assumed plan: errors in x_{ref} propagate into a wrong $s(t)$ and degrade cost even when terminal accuracy is preserved. An online-estimated or predicted reference inherits this sensitivity.

No replanning. The finite-horizon Riccati pair is computed once for a fixed $[0, t_f]$ and a fixed plan; it cannot react to a reference that is revealed, revised, or re-targeted online.

These limitations motivate causal alternatives that the present formulation does not provide: a receding-horizon (model-predictive) reformulation that re-solves the tracking problem over a moving window as new reference information arrives, a causal reference model/observer that predicts x_{ref} ahead, or a feedforward driven by an online reference generator. Each restores anticipation at the cost of repeated online optimization.

4.6. Validation

Because the Part II objective has no terminal cost and the final state is free, optimality is certified against the PMP necessary conditions rather than terminal convergence. The tracking Riccati pair satisfies the homogeneous free-terminal conditions $P(t_f) = 0$ and $s(t_f) = 0$, and the stationarity condition $Ru + B^\top \lambda = 0$ (with costate $\lambda = P(t)e(t) + s(t)$) holds along the entire closed-loop rollout. Table 7 reports these residuals at the level of machine precision. The terminal tracking error is *not* required to vanish for this free-final-state problem; its value ($\|e(t_f)\|_\infty \approx 1.16$, dominated by the vertical/mass relaxation analyzed above) is a result, not a residual.

Table 7: Part II Tracking: PMP Optimality Residuals

Metric	Target	Computed	Error (Residual)
$\ P(t_f)\ _\infty$ (Riccati terminal)	0	1.0×10^{-14}	1.0×10^{-14}
$\ s(t_f)\ _\infty$ (feedforward terminal)	0	1.6×10^{-17}	1.6×10^{-17}
$\max_t \ Ru + B^\top \lambda\ $ (stationarity)	0	1.1×10^{-14}	1.1×10^{-14}

The state-equation defect of the RK4 rollout is $\mathcal{O}(10^{-1})$ at the chosen step. Figure 18 shows the Hamiltonian evolution: for the autonomous reference plan H is constant to $\sim 10^{-4}$, whereas for the tracking problem H is intentionally time-varying because the explicit dependence on $x_{\text{ref}}(t)$ makes $\partial H / \partial t \neq 0$ – the relevant optimality certificate for tracking is the stationarity/costate residual rather than constancy of H .

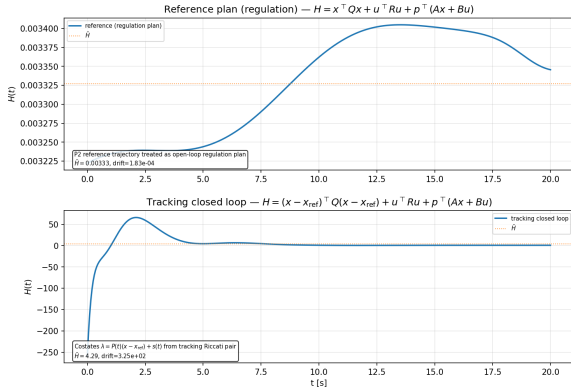


Figure 18: Hamiltonian consistency: constant for the autonomous reference plan and time-varying for the tracking problem.

5. PART III - The Mission Problem

5.1. Phase A — Minimum-Time Ascent

Formulation.

$$J_A = \int_0^{t_1} 1 dt = t_1$$

Analytical Derivations (PMP). Hamiltonian:

$$H_A(x, u, p) = 1 + p^\top (Ax + Bu) = 1 + p^\top Ax + p^\top Bu$$

Adjoint equation:

$$\dot{p} = -\frac{\partial H_A}{\partial x} = -A^\top p$$

Switching Function:

$$\frac{\partial H_A}{\partial u} = B^\top p \implies S(t) = B^\top p(t)$$

$$u_i^*(t) = \begin{cases} U_{\max,i} & \text{if } S_i(t) < 0 \\ U_{\min,i} & \text{if } S_i(t) > 0 \end{cases}$$

$$H_A(t) \equiv 0 \implies 1 + p(t)^\top (Ax(t) + Bu^*(t)) = 0 \quad \forall t \in [0, t_1]$$

Implementation.

Ascent Trajectory Results. Why thrust saturation appears.

How mass depletion changes control authority during flight.

How the switching structure evolves over time.

Whether the optimal strategy front-loads or back-loads energy consumption.

Note that the behavior largely depends on the choice of x_{sky} . Choose the terminal velocity to be zero versus it is positive/negative and compare.

Validation.

5.2. Phase B — Return-and-Land Problem

Phase B begins precisely where Phase A ends, thus the initial condition for the landing phase is $x(0) = x_1$.

Formulation & Core PMP.

$$J_B = \int_0^{t_f} (x^\top Qx + u^\top Ru) dt$$

Generic Hamiltonian, and the fundamental $H(t_f) = 0$ condition:

$$H_B(x, u, p) = x^\top Qx + u^\top Ru + p^\top (Ax + Bu)$$

The unconstrained control law and adjoint equations remain identical to Phase I:

$$u^*(t) = -\frac{1}{2} R^{-1} B^\top p(t)$$

$$\dot{p}(t) = -2Qx(t) - A^\top p(t)$$

Because t_f is free, PMP imposes the free-time transversality condition. The Hamiltonian evaluated at the terminal time must equal zero:

$$H_B(t_f) = 0 \implies x(t_f)^\top Qx(t_f) + u(t_f)^\top Ru(t_f) + p(t_f)^\top (Ax(t_f) + Bu(t_f)) = 0$$

Transversality.

$$p(t_f) = \frac{\partial \phi}{\partial x} \Big|_{t_f} + \sum_j v_j \nabla h_j(x(t_f))$$

Manifold 1 (Pinpoint Landing)

Description

$$h_1(x) = p_z = 0$$

$$h_2(x) = v_z = 0$$

$$h_3(x) = \theta = 0$$

Gradients

$$\nabla h_1 = [0, 1, 0, 0, 0, 0, 0]^\top$$

$$\nabla h_2 = [0, 0, 0, 1, 0, 0, 0]^\top$$

$$\nabla h_3 = [0, 0, 0, 0, 1, 0, 0]^\top$$

Transversality

$$p_{p_x}(t_f) = \frac{\partial \phi}{\partial p_x} \quad (\text{Free state})$$

$$p_{p_z}(t_f) = \frac{\partial \phi}{\partial p_z} + v_1 \quad (\text{Constrained})$$

$$p_{v_x}(t_f) = \frac{\partial \phi}{\partial v_x} \quad (\text{Free state})$$

$$p_{v_z}(t_f) = \frac{\partial \phi}{\partial v_z} + v_2 \quad (\text{Constrained})$$

$$p_\theta(t_f) = \frac{\partial \phi}{\partial \theta} + v_3 \quad (\text{Constrained})$$

$$p_\omega(t_f) = \frac{\partial \phi}{\partial \omega} \quad (\text{Free state})$$

$$p_m(t_f) = \frac{\partial \phi}{\partial m} \quad (\text{Free state})$$

Manifold 2 (Surface Landing)

Description: landing on the surface of a dome/sphere.

$$h_1(x) = (p_x - p_c)^2 + p_z^2 - r^2 = 0$$

$$h_2(x) = v_z = 0$$

$$h_3(x) = \theta = 0$$

The Gradients: h_1 couples p_x and p_z , spanning 2 dimensions.

$$\nabla h_1 = [2(p_x - p_c), 2p_z, 0, 0, 0, 0, 0]^\top$$

$$\nabla h_2 = [0, 0, 0, 1, 0, 0, 0]^\top$$

$$\nabla h_3 = [0, 0, 0, 0, 1, 0, 0]^\top$$

Transversality

$$p_{p_x}(t_f) = \frac{\partial \phi}{\partial p_x} + 2v_1(p_x(t_f) - p_c)$$

$$p_{p_z}(t_f) = \frac{\partial \phi}{\partial p_z} + 2v_1 p_z(t_f)$$

(The remaining 5 costates are identical to M1)

Conclusion for M2: We can eliminate the unknown multiplier v_1 by dividing the first two equations (assuming $\frac{\partial \phi}{\partial x} = 0$). This yields the exact boundary condition passed to the TP-BVP solver:

$$\frac{p_{p_x}(t_f)}{p_{p_z}(t_f)} = \frac{p_x(t_f) - p_c}{p_z(t_f)}$$

Landing Manifold Comparison.

Landing time

Energy consumption

Switching structure

Terminal sensitivity

Feasibility

How the geometry of the terminal manifold affects the resulting optimal trajectory and control structure.

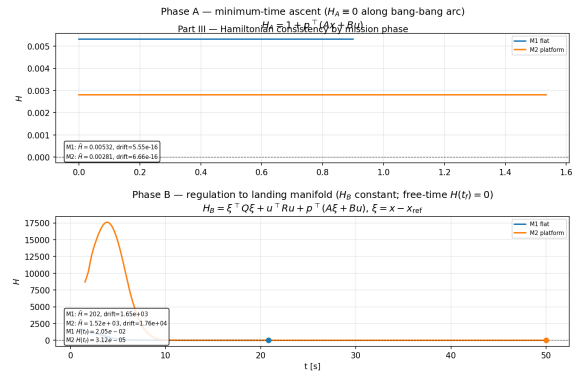


Figure 19: Hamiltonian evolution

Validation.

6. PART IV - The Nonlinear Optimal Control Near Hover

6.1. Phase Formulation

Consider the fixed-time, free-endpoint nonlinear optimal control problem:

$$\min_u \int_0^{t_f} (x^\top Qx + u^\top Ru) dt$$

Recall: State Equations (which is also $\dot{x} = \frac{\partial H}{\partial p}$):

$$\dot{x} = \begin{bmatrix} v_x \\ v_z \\ -\frac{T}{m} \sin \theta \\ \frac{T}{m} \cos \theta - g \\ \omega \\ \frac{\tau}{I} \\ -\alpha T \end{bmatrix}$$

System is anchored by the known initial launch state at $t = 0$:

$$x(0) = x_0$$

6.2. Analytical Derivations

Nonlinear Hamiltonian

Adjoin the nonlinear dynamics to the running cost using the costate vector $p = [p_1, p_2, p_3, p_4, p_5, p_6, p_7]^\top$:

$$H(x, u, p) = x^\top Qx + u^\top Ru + p_1 v_x + p_2 v_z - p_3 \frac{T}{m} \sin \theta + p_4 \left(\frac{T}{m} \cos \theta - g \right) + p_5 \omega + p_6 \frac{\tau}{I} - p_7 \alpha T$$

Adjoint equations (costate equations)

$$\dot{p} = -\frac{\partial H}{\partial x}$$

$$\dot{p}_1 = -2Q_{11}p_x$$

$$\dot{p}_2 = -2Q_{22}p_z$$

$$\dot{p}_3 = -2Q_{33}v_x - p_1$$

$$\dot{p}_4 = -2Q_{44}v_z - p_2$$

$$\dot{p}_5 = -2Q_{55}\theta + p_3 \frac{T}{m} \cos \theta + p_4 \frac{T}{m} \sin \theta$$

$$\dot{p}_6 = -2Q_{66}\omega - p_5$$

$$\dot{p}_7 = -2Q_{77}m - p_3 \frac{T}{m^2} \sin \theta + p_4 \frac{T}{m^2} \cos \theta$$

Stationarity conditions ($\frac{\partial H}{\partial u} = 0$)

Consistent with the implementation and with Parts I–II, the running cost penalizes the *deviation* control $u = [\delta T, \tau]^\top$, where $\delta T = T - m_0 g$ is the thrust increment about the hover trim and the thrust

entering the equations of motion is $T = m_0 g + \delta T$ (so $\partial T / \partial \delta T = 1$). Differentiating H directly with respect to the deviation control yields the optimal increment, with no trim offset subtracted afterwards:

For the thrust increment (δT):

$$\frac{\partial H}{\partial \delta T} = 2R_{11} \delta T - p_3 \frac{\sin \theta}{m} + p_4 \frac{\cos \theta}{m} - p_7 \alpha = 0$$

$$\delta T^* = \text{sat}_{\delta T_{\min}}^{\delta T_{\max}} \left[\frac{1}{2R_{11}} \left(p_3 \frac{\sin \theta}{m} - p_4 \frac{\cos \theta}{m} + p_7 \alpha \right) \right],$$

where the saturation limits are the actuator bounds expressed in deviation form, $\delta T \in [T_{\min} - m_0 g, T_{\max} - m_0 g]$.

For Torque (τ):

$$\frac{\partial H}{\partial \tau} = 2R_{22} \tau + \frac{p_6}{I} = 0$$

$$\tau^* = \text{sat}_{\tau_{\min}}^{\tau_{\max}} \left[-\frac{p_6}{2IR_{22}} \right]$$

Transversality (terminal condition)

Because the terminal state is subject to optimization, the terminal costate at $t = t_f$ is governed by the transversality condition based on the terminal penalty $\phi(x(t_f))$ (or specific manifold constraints):

$$p(t_f) = \frac{\partial \phi}{\partial x} \Big|_{t=t_f}$$

Complete TPBVP

The complete TPBVP is formed by substituting the optimal, saturated control laws (δT^* , τ^*) back into the 7 state equations and 7 adjoint equations. This creates a deeply coupled system of 14 nonlinear, first-order ordinary differential equations. This system must be solved simultaneously across the domain $t \in [0, t_f]$, subject to the split boundary conditions $x(0)$ and $p(t_f)$.

$$\begin{aligned}
\dot{p}_x &= v_x \\
\dot{p}_z &= v_z \\
\dot{v}_x &= -\frac{T}{m} \sin \theta \\
\dot{v}_z &= \frac{T}{m} \cos \theta - g \\
\dot{\theta} &= \omega \\
\dot{\omega} &= \frac{\tau}{I} \\
\dot{m} &= -\alpha T \\
\dot{p}_1 &= -2Q_{11}p_x \\
\dot{p}_2 &= -2Q_{22}p_z \\
\dot{p}_3 &= -2Q_{33}v_x - p_1 \\
\dot{p}_4 &= -2Q_{44}v_z - p_2 \\
\dot{p}_5 &= -2Q_{55}\theta + \frac{T}{m}(p_3 \cos \theta + p_4 \sin \theta) \\
\dot{p}_6 &= -2Q_{66}\omega - p_5 \\
\dot{p}_7 &= -2Q_{77}m + \frac{T}{m^2}(p_4 \cos \theta - p_3 \sin \theta)
\end{aligned} \tag{1}$$

6.3. Numerical Implementation

The Part IV problem is solved in hover-deviation coordinates (x is the deviation from the hover trim $T^* = m_0g$, $\theta^* = 0$; control $u = [\delta T, \tau]^\top$), so that the linearized solutions of the previous parts can serve directly as the nonlinear warm start. The implementation (`nonlinear.py`) builds three comparison scenarios on a common initial condition x_0 and a common short horizon:

Restriction to the near-hover regime. Because true hover is *not* an equilibrium of a variable-mass vehicle (holding $T = mg$ continuously depletes mass at $\dot{m} = -\alpha mg$), the horizon is kept short, $t_f = 4$ s with 201 uniform nodes. This keeps the trajectory near hover and the mass safely above m_{dry} , which is the validity regime prescribed for this part.

Scenario 1 — LQR on the linear plant. The backward Riccati sweep of Part I produces $P(t)$ and the feedback $u = -R^{-1}B^\top P(t)x$, integrated forward on $\dot{x} = Ax + Bu$.

Scenario 2 — LQR on the nonlinear plant. The *same* time-varying gain is applied to the full nonlinear EOM, integrated with a fixed-step RK4 scheme, with control saturation enforced by clipping. This isolates the modeling error incurred by using the linear law on the true plant.

Scenario 3 — Nonlinear PMP / TPBVP.

The coupled 14-state state/costate system of Eq. (1) is solved by collocation (`scipy.integrate.solve_bvp`). The optimal control is substituted in closed form from the stationarity conditions and saturated to the actuator bounds. The split boundary conditions are the initial-state match $x(0) = x_0$ and the free-endpoint transversality $p(t_f) = 2Q_f x(t_f)$.

Warm start and costate scaling. The collocation guess uses the Scenario 1 state history and the costate guess $p(t) = 2P(t)x(t)$; the factor of two maps the $\frac{1}{2}$ -scaled Riccati costate to the no- $\frac{1}{2}$ running cost $x^\top Qx + u^\top Ru$ used here. Convergence is declared at a collocation tolerance of 10^{-6} .

6.4. Analysis

The following 3 simulation scenarios will be compared:

- linear optimal controller on the linearized dynamics
- the same controller on the nonlinear dynamics
- nonlinear optimal controller on the nonlinear dynamics.

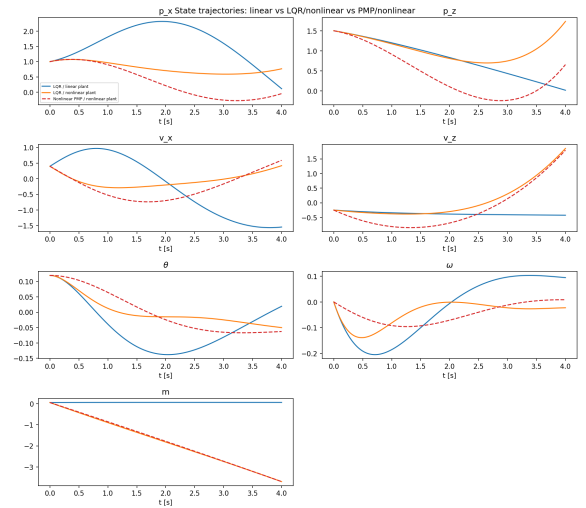


Figure 20: Trajectory comparisons (position, velocity, attitude, and mass trajectories)

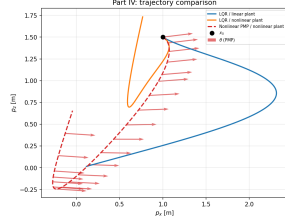


Figure 21: Trajectory comparisons (px vs pz)

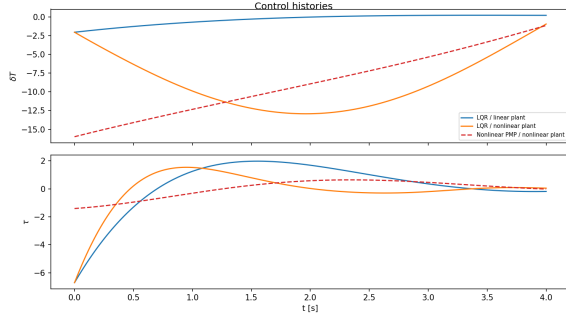


Figure 22: Control histories (thrust and torque histories)

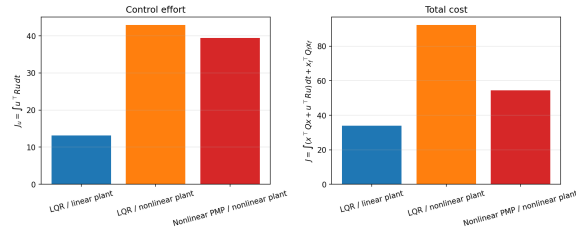


Figure 23: Energy consumption: control effort $J_u = \int u^T R u dt$ and total cost J for the three scenarios.

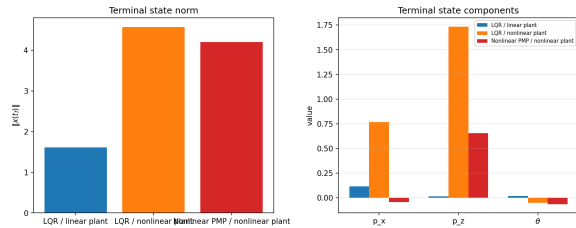


Figure 24: Terminal error comparisons (terminal state norm and components).

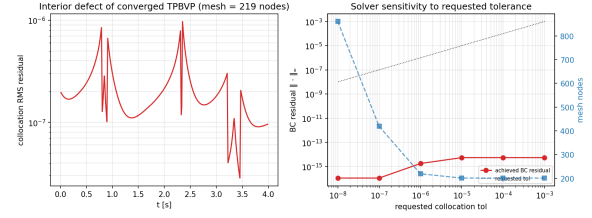


Figure 25: Convergence behavior: interior collocation residual of the converged TPBVP and solver sensitivity of the achieved boundary residual / mesh size to the requested tolerance.

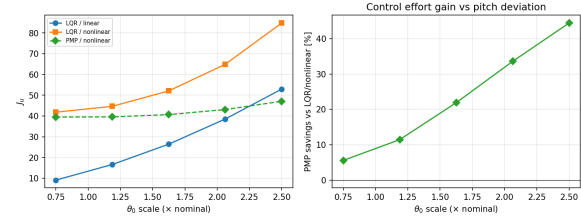


Figure 26: Solver sensitivity: control effort and PMP-vs-LQR savings as the initial pitch deviation θ_0 is scaled.

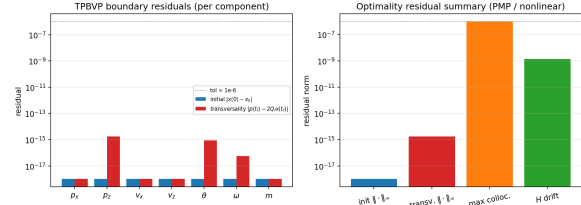


Figure 27: Terminal residual analysis: per-component TPBVP boundary residuals (initial match and free-endpoint transversality $p(t_f) = 2Q_f x(t_f)$) and the optimality-residual summary.

6.5. Validation

To verify the numerical solution to the Two-Point Boundary Value Problem (TPBVP) and prove strict adherence to Pontryagin's Maximum Principle (PMP), the solution was validated against the necessary optimality conditions.

Hamiltonian Consistency. Part IV is a *fixed*-final-time, free-endpoint problem, and the hover-trim dynamics are autonomous. The Hamiltonian must therefore remain *constant* along the optimal trajectory; unlike the free-final-time mission of Part III, this constant is *not* required to vanish. As shown in Figure 28, the Hamiltonian of the nonlinear PMP solution is constant to a drift of $\mathcal{O}(10^{-9})$ about $H(t_f) \approx 34.7$, confirming Hamiltonian consistency. By contrast, applying the linear LQR

law to the nonlinear plant produces a Hamiltonian that drifts by $\mathcal{O}(10^2)$ over the same horizon—a direct, quantitative certificate that the linear law is *not* an extremal of the nonlinear problem.

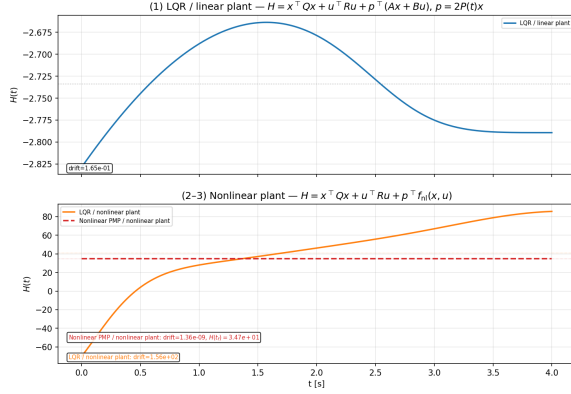


Figure 28: Hamiltonian evolution $H(t)$ for the three scenarios. The nonlinear PMP solution (bottom) holds H constant to $\mathcal{O}(10^{-9})$ as required for an autonomous fixed-time extremal, whereas the LQR law on the nonlinear plant drifts by $\mathcal{O}(10^2)$.

Terminal Conditions, Adjoint Boundaries, and Shooting Residuals. Because the terminal state is free, the boundary conditions verified are the initial-state match $x(0) = x_0$ and the free-endpoint transversality $p(t_f) = 2Q_f x(t_f)$; there are no fixed terminal-state targets in this part. Table 8 reports the per-component boundary residuals together with the interior collocation defect and the Hamiltonian drift.

The global $\mathcal{L}_\infty$ norm of the boundary-condition residual was 1.8×10^{-15} and the maximum interior collocation residual was 9.7×10^{-7} on a converged mesh of 219 nodes—both within the requested solver tolerance, confirming that the computed solution satisfies the necessary optimality conditions to numerical precision.

Table 8: Part IV validation: boundary and optimality residuals of the nonlinear TPBVP (fixed-time, free endpoint).

Type	Boundary / optimality condition	Residual
Initial	$x(0) = x_0$ ($\mathcal{L}_\infty$)	3.9×10^{-26}
Transv.	$p_{p_z}(t_f) = 2Q_{f,22} p_z(t_f)$	1.8×10^{-15}
Transv.	$p_\theta(t_f) = 2Q_{f,55} \theta(t_f)$	8.9×10^{-16}
Transv.	$p_\omega(t_f) = 2Q_{f,66} \omega(t_f)$	5.6×10^{-17}
Transv.	$p(t_f) = 2Q_f x(t_f)$ ($\mathcal{L}_\infty$)	1.8×10^{-15}
Hamiltonian	$H(t)$ drift (autonomous, fixed t_f)	1.4×10^{-9}
Collocation	interior RMS defect (219 nodes)	9.7×10^{-7}

ACKNOWLEDGMENT

The author thanks Professor Shahriar Talebi for guidance and instruction throughout the MAE 270C Optimal Control course. Their insights into optimal control theory formed the foundation of this project.

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